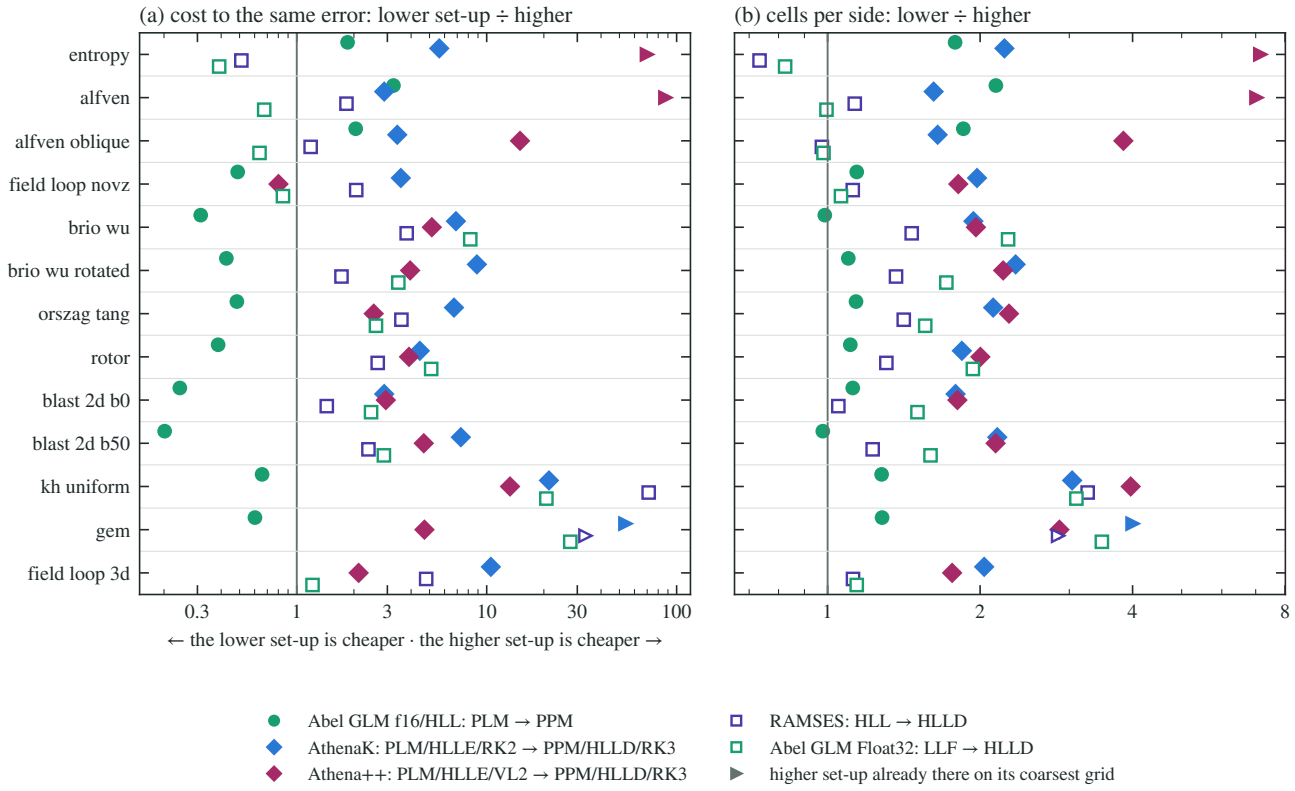
(a) cost to the same error: lower set-up ÷ higher
(b) cells per side: lower ÷ higher
entropy
alfven
alfven oblique
field loop novz
brio wu
brio wu rotated
orszag tang
rotor
blast 2d b0
blast 2d b50
kh uniform
gem
field loop 3d
0.3
1
3
10
30
100
1
2
4
8
← the lower set-up is cheaper · the higher set-up is cheaper →
Abel GLM f16/HLL: PLM → PPM
AthenaK: PLM/HLLE/RK2 → PPM/HLLD/RK3
Athena++: PLM/HLLE/VL2 → PPM/HLLD/RK3
RAMSES: HLL → HLLD
Abel GLM Float32: LLF → HLLD
higher set-up already there on its coarsest grid